

Dynamic Rotation-Stacked Visibility Graph (DRVG)

Abstract

1 Convex Case

Let $W \subset \mathbb{R}^2$ be a compact polygonal workspace with finitely many polygonal obstacles $\mathcal{O}$, and let R be a polygonal robot whose reference point c is its center. Assume that R is star-shaped with respect to c . Fix a rotational resolution n large enough that the corresponding fully known RVG planner is resolution-complete under its usual clearance assumptions.

We aim to prove that if a feasible path from q_s to q_g exists in the component covered by the resolution-completeness hypothesis, the online planner returns a feasible path from q_s to q_g after finitely many primitive moves.

Let R_θ denote the footprint of the robot at orientation θ , centered at the origin. Since R is star-shaped with respect to its center, for every $p \in R_\theta$ and every $\lambda \in [0, 1]$, we have $\lambda p \in R_\theta$.

First, consider a collision-free straight translation at fixed orientation θ , moving the center from a to b . For any point $b + p$ of the final footprint, where $p \in R_\theta$, the segment from a to $b + p$ is

$$\gamma(\lambda) = a + \lambda(b - a + p) = a + \lambda(b - a) + \lambda p, \quad \lambda \in [0, 1].$$

Because $\lambda p \in R_\theta$, the point $\gamma(\lambda)$ lies in the robot footprint centered at $a + \lambda(b - a)$. Hence the entire segment lies in the swept volume of the translation. Since the translation is collision-free, this segment is obstacle-free. Therefore the final footprint is visible from the current center.

For an in-place rotation, the center is fixed. If the target footprint $a + R_{\theta'}$ is collision-free, then for every $p \in R_{\theta'}$ and $\lambda \in [0, 1]$,

$$a + \lambda p \in a + R_{\theta'}.$$

Thus every point of the target footprint is visible from the center a . Consequently, every primitive move made by the planner can be certified using visible free space.

Second, any path may be represented in a normal form with no two consecutive rotation primitives. Indeed, two consecutive in-place rotations occur about the same center a and form a single continuous path in the fiber $\{a\} \times SO(2)$. Replacing their concatenation by one rotation primitive with the same orientation trace preserves the endpoint, swept set, and feasibility. Hence it suffices to consider paths that alternate between translations and rotations.

Third, consider a maximal straight translation in one RVG layer of the currently discovered free space. In a polygonal visibility graph, such a move can end only at the goal, at a vertex of a grown obstacle or workspace boundary, or at a frontier vertex. The first case terminates successfully. The second case can occur only finitely many times, since the workspace, obstacles, robot, and resolution n induce only finitely many such graph states, and each state is expanded at most once by assumption. In the third case, the endpoint is a dynamic frontier vertex. Since we perform Minkowski sums over the polygonal visibility graphs, this is equivalent to a point visibility graph, and we will thus always reveal new area.

After finitely many such moves, either the goal has been found, or no reachable dynamic frontier vertex remains. Suppose, for contradiction, that some undiscovered free region in the start component remains reachable. Let τ be a feasible path from the discovered region into that undiscovered region, and let t^* be the first parameter at which τ leaves the discovered region. Then $\tau(t^*)$ lies on a reachable frontier. By assumption, this frontier induces a reachable dynamic frontier vertex, contradicting the fact that none remain. Hence the entire reachable component has been discovered.

Once this component is fully discovered, the online RVG is the same, up to irrelevant duplicate vertices and tie-breaking choices, as the RVG that would have been constructed if the environment had been known in advance. Since n is large enough for the fully known RVG planner to be resolution-complete, a graph path from q_s to q_g exists whenever a feasible path exists under the same clearance assumptions. Therefore the online planner must find and return such a path after finitely many primitive moves.

References